

Microeconomic Analysis: Problem Set II

Multivariate Calculus

Due in week 3:

Please submit questions 5, 7 and 10.
Bring all the other questions to the class.

1. **Properties of Numbers.** Show that there is no rational number $a = p/q$ (in its simplest form) such that $a^2 = 2$. Use this fact to show this $\mathbb{Q}$ is not complete.

Hint: Argue by contradiction. Assume that $p^2/q^2 = 2$ and show that both p and q are even.

2. **Norms.** The Cauchy-Schwarz inequality states that $|\sum_{i=1}^n x_i y_i| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ (where $\|\cdot\|$ represents the Euclidean norm). Use the Cauchy-Schwarz inequality to show that $\|x + y\|^2 \leq (\|x\| + \|y\|)^2$ and hence prove the Triangle Inequality for the Euclidean norm.

3. **Limits.** Using the algebra of limits, evaluate the following limits:

$$(a) \lim_{x \rightarrow \infty} \frac{x-2}{3x^2+2}, \quad (b) \lim_{x \rightarrow \infty} \frac{x^2+1}{4x^2+x}, \quad (c) \lim_{x \rightarrow 1+} \frac{1}{(x-1)(x-2)}.$$

4. **Sets.**

- (a) For each of the following subsets of $\mathbb{R}$, say whether the set is (i) open, closed, or neither; (ii) bounded; (iii) compact; (iv) convex; and write down the supremum, and the maximum if it exists.

$$A = (0, 1), \quad B = (-\infty, -1], \quad C = \{x : |x| > 2\}, \quad \mathbb{R} \setminus C.$$

- (b) Let $A = \{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$. Sketch the set, and say whether it is (i) open, closed or neither; (ii) bounded; (iii) compact; (iv) convex.
- (c) Show that if $\{A_i \subset \mathbb{R}^n : i \in I\}$ is a collection of open sets, $\cup_{i \in I} A_i$ is an open set. Analogously, show that is $\{A_i \subset \mathbb{R}^n : i = 1, \dots, m\}$ is a finite collection of open sets, $\cap_{i=1}^m A_i$ is an open set.

5. Continuity.

- (a) Show using the (ϵ, δ) definition of continuity that $f(x, y) = xy$ is continuous at $(0, 0)$.
- (b) Show using the (ϵ, δ) definition of continuity that if $f : \mathbb{R}^m \rightarrow \mathbb{R}$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}$ are continuous at $\mathbf{x}_0$, then $f + g$ is continuous at $\mathbf{x}_0$.
- (c) Prove that if $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is continuous then, if A is an open set in $\mathbb{R}^m$, $f^{-1}(A)$ is open in $\mathbb{R}^n$.
Hint: write the definition of continuity in terms of inclusions of open balls.

6. **Fixed point.** The function g is defined by $g(x) = 2x(2 - x)$, for $x \in [0, 2]$. Show that Brouwer's fixed point theorem applies, and find its fixed points.

7. **Differentiation.** Consider two functions

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{otherwise} \end{cases}$$

$$g(x, y) = \begin{cases} x^{\frac{1}{2}}y^{\frac{1}{2}} & \text{if } x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Use the formal definition to compute the partial derivatives of the functions at $(0, 0)$.
- (b) Are the functions of class C^1 ?
- (c) Are the functions differentiable at $(0, 0)$?

8. Cobb-Douglas Utility

Consider a Cobb-Douglas utility function: $u = \sum_{i=1}^L \alpha_i \ln x_i$ with $\alpha_i > 0$ and $\sum_{i=1}^L \alpha_i = 1$.

- (a) Compute the Jacobian Du .
- (b) Compute the Hessian D^2u .
- (c) Show that D^2u is symmetric and negative definite using the determinant test.
- (d) Show that D^2u is negative definite by directly checking the sign of the quadratic form $\mathbf{y}^T D^2u \mathbf{y}$.
- (e) Provide a second-order Taylor approximation of $u(\cdot)$ around $\mathbf{x} = (1, \dots, 1)^T$.

9. Implicit Function I

Consider the equation $z^2 + xz + yx^2 + y^3 = 0$ and the solution $s^* = (x^*, y^*, z^*) = (1, 0, 0)$.

- (a) Show that the equation defines a function $z = g(x, y)$ in a neighborhood of $(1, 0)$.

- (b) Compute the partial derivatives of this function z at $(1, 0)$.

10. Implicit Function II

Consider the system of equations:

$$\begin{aligned}x^2 - y^2 - u^3 + v^2 + 4 &= 0 \\ 2xy + y^2 - 2u^2 + 3v^4 + 8 &= 0\end{aligned}$$

and the solution $s^* = (x^*, y^*, u^*, v^*) = (2, -1, 2, 1)$.

- (a) Show that you can express the values of (u, v) that solve the equations as a function of the values of (x, y) , in a neighborhood of $(2, -1)$
- (b) At $(2, -1)$, how do u and v change when x changes?